PL Insights: Data Quality Validation Report

Course: DL3004 Data Warehousing & Business Intelligence | Phase 2 Implementation | Date: November 2026

1. Data Quality Framework Overview

A rigorous, multi-tier data quality framework was implemented directly in PostgreSQL (project_dw.dq_results) to validate the structural and mathematical soundness of the Data Warehouse. The suite evaluates 30 test cases spanning **Completeness**, **Accuracy (Reconciliation)**, **Referential Integrity**, and **Volume Benchmarks**. All 28 critical assertions passed successfully (0 failures, 2 informational tracking metrics).

Total Assertions	Passed Assertions	Failed Assertions	Informational Checks	Overall Status
30 Checks	**28 (100% Critical)**	**0 (Zero Failures)**	**2 (Tracking Unknowns)**	**PASSED (Production Ready)**

2. Comprehensive Data Quality Test Matrix (project_dw.dq_results)

ID	Category	Assertion / Check Name	Expected	Actual	Status
1	**Completeness**	fact_match_team NULL date_sk	0	**0**	**PASS**
2	**Completeness**	fact_match_team NULL season_sk	0	**0**	**PASS**
3	**Completeness**	fact_match_team NULL team_sk	0	**0**	**PASS**
4	**Completeness**	fact_match_team NULL opponent_team_sk	0	**0**	**PASS**
5	**Completeness**	fact_match_team NULL referee_sk	0	**0**	**PASS**
6	**Completeness**	fact_match_team NULL venue_sk	0	**0**	**PASS**
7	**Completeness**	fact_match_team NULL goals_for/against	0	**0**	**PASS**
8	**Completeness**	dim_team NULL team_name	0	**0**	**PASS**
9	**Completeness**	dim_referee NULL referee_name	0	**0**	**PASS**
10	**Completeness**	dim_season NULL season_code	0	**0**	**PASS**
11	**Completeness**	dim_date NULL full_date (excl. unknown)	0	**0**	**PASS**
12	**Completeness**	fact rows with unknown referee (-1) [info]	n/a	**0**	**INFO**
13	**Completeness**	fact rows with unknown venue (-1) [info]	n/a	**0**	**INFO**
14	**Accuracy**	SUM(goals_for) in DW = SUM(FTHG + FTAG) in staging	17018	**17018**	**PASS**
15	**Accuracy**	fact_match_team count = 2 x clean matches (12,160)	12160	**12160**	**PASS**
16	**Accuracy**	seasons with != 380 matches	0	**0**	**PASS**
17	**Accuracy**	team-seasons with != 38 matches	0	**0**	**PASS**
18	**Accuracy**	matches with points not summing to 3 (win/loss) or 2 (draw)	0	**0**	**PASS**
19	**Integrity**	fact rows without dim_date	0	**0**	**PASS**
20	**Integrity**	fact rows without dim_season	0	**0**	**PASS**
21	**Integrity**	fact rows without dim_team	0	**0**	**PASS**
22	**Integrity**	fact rows without opponent dim_team	0	**0**	**PASS**
23	**Integrity**	fact rows without dim_referee	0	**0**	**PASS**
24	**Integrity**	fact rows without dim_venue	0	**0**	**PASS**
25	**Integrity**	snapshot rows without dim_team	0	**0**	**PASS**
26	**Integrity**	factless rows without dim_season	0	**0**	**PASS**
27	**Integrity**	duplicate grain keys (match_id, team_sk)	0	**0**	**PASS**
28	**Volume**	fact_match_team rows >= 10,000 (Rubric Req)	>=10000	**12160**	**PASS**
29	**Volume**	dim_date records >= 100 (Rubric Req)	>=100	**5845**	**PASS**
30	**Volume**	dim_team records >= 40 (Across 16 seasons)	>=40	**42**	**PASS**

3. Category Findings & Mathematical Validations

• **Completeness (Checks 1-13):** Verified zero NULL values across all primary keys, foreign keys, and statistical measures in fact_match_team, dim_date, dim_season, dim_team, and dim_referee. Informational checks confirm 0 records required fallback to unknown referee (-1) or unknown venue (-1).
• **Accuracy & Reconciliation (Checks 14-18):** Reconciliation confirmed that SUM(goals_for) in the dimensional warehouse (17,018) exactly equals SUM(FTHG + FTAG) in the raw staging layer (17,018). Exactly 12,160 fact records exist (2 x 6,080 matches). All 16 seasons possess exactly 380 matches, every team-season contains 38 matches, and 100% of matches sum to 3 points (decisive win/loss) or 2 points (draw).
• **Referential Integrity (Checks 19-27):** Full outer/left anti-joins between fact tables and all dimensional tables returned exactly zero orphan records. Unique grain constraint on (match_id, team_sk) confirmed zero duplicate entries.
• **Volume Standards (Checks 28-30):** Dimensional warehouse strictly satisfies all grading criteria: fact_match_team holds 12,160 rows (exceeding the 10,000 threshold), dim_date holds 5,845 calendar days (exceeding the 100 threshold), and dim_team accounts for 42 entries (41 clubs + 1 unknown member).